Enabling Change Data Capture on SAP BDC Data Products

What the standard configuration teaches, and what to ask for on a custom one

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Vendor-neutral community pattern | Apache 2.0 | All measurements from live SAP BDC tenants

Standard SAP BDC data products ship four connector-generated change columns. Custom data products expose only what their pipeline owner chose to. This paper treats the standard configuration as a specification, scores a well-configured custom product against it, and arrives at a recommendation that is not the obvious one: on the two properties that govern cost and correctness the custom track measured ahead, and the highest-value action requires nothing from SAP at all.

1. The question, and the short answer

Standard SAP BDC data products arrive with four connector-generated change columns. Custom data products - those built through SLT configuration - arrive with whatever their pipeline owner chose to expose, commonly a single change timestamp. The question this paper answers is what a custom data product should carry in order to support incremental capture properly, using the standard configuration as the specification.

The short answer is not the obvious one. Of the four standard columns, one should be adopted, one should be adopted but NOT in the form SAP ships it, one is a convenience, and one has no value for capture at all. On the two properties that actually determine what capture costs and whether it is correct - watermark typing and predicate pruning - the reference custom product measured ahead of the reference standard product.

> And before asking SAP for anything: the highest-value action found in this work required nothing from SAP. The reference custom share already carried SAP's own deletion and blocking codes, populated on 19.0% of purchase order items and accounting for 42.2% of reported USD ordered value, and the Gold layer discarded them. Section 5.

2. Why the standard configuration is the right yardstick

A catalog-linked database holds Iceberg tables on an SAP-managed external catalog. Snowflake's three native change mechanisms are all refused on them, which is what makes the share's own columns the whole of the available design space:

Attempted operation	Error returned
CREATE STREAM ... ON TABLE <share>	Streams on External Tables or Iceberg tables with an external catalog must have INSERT_ONLY set to true
SELECT ... CHANGES(INFORMATION => DEFAULT)	CHANGES clause is not supported on external tables or Iceberg tables with an external catalog
ALTER TABLE ... SET CHANGE_TRACKING = TRUE	change_tracking cannot be set for Iceberg tables unless they use Snowflake catalog

SHOW TABLES reports change_tracking = ON on these objects regardless. That flag is uninformative here and must not be read as evidence that a stream-based design is available.

Because nothing can be added from the Snowflake side, whatever the share carries is the change signal. The four standard columns are therefore worth reading as SAP's own answer to the question 'what does a consumer need in order to process this incrementally' - not as a guess, but as a shipped contract. That makes them the natural specification to hold a custom product against.

3. What each standard column is actually worth

Measured on a live standard data product across five non-empty objects (67,480 rows). The verdict column is the point: the four columns are not of equal value, and one of them has no value for capture at all.

Column	What was measured	Verdict
__TIMESTAMP	Row write time. TEXT, ISO-8601 with a COMMA decimal separator (2026-06-15T09:41:59,6426580Z). MAX() on the raw text scanned 0.000 MB in 2 ms from Iceberg manifest statistics; MAX() over a parsed expression scanned 0.310 MB in 45 ms. Lexicographic order equals chronological order, verified.	Essential. The TEXT typing is a liability, not a feature.
__OPERATION_TYPE	Change mode. Only L (initial load, 67,087 rows) and A (delta after-image, 393 rows) ever appeared. No delete code was observed anywhere, and the authoritative value list could not be confirmed from SAP documentation.	Necessary in principle. Unproven in practice.
load_type_<hash>	Load phase. initial = 67,087, delta = 393. Confirms the connector appends rather than reloading, which is the precondition for any watermark to mean anything.	Useful, cheap, and replaceable by inference.
run_id_<hash>	Held the SAME value on the initial snapshot and on every delta that followed it. Identifies the replication flow, not the batch. The 32-hex suffix is likewise per product, not per object - identical across all seven objects.	No value for capture. Do not replicate it.

The __TIMESTAMP finding is the one that generalises into a rule. Because the format is fixed-width and zero-padded, a text comparison is both correct and free, while parsing before comparing turns a free poll into a scan on every cycle in order to learn nothing. Any implementation on this track should compare as text and parse only for display.

4. Scoring a custom product against the specification

The reference custom-track account carries ZZLASTCHGDATE on 23 of 23 L1 objects, with 59 objects carrying it account-wide and no connector metadata columns anywhere. Scoring it against the four standard capabilities, plus the two properties that govern cost:

Capability	Standard product	Custom product	Ahead
Row change timestamp	__TIMESTAMP, TEXT with comma separator; an unguarded cast returns NULL on every row	ZZLASTCHGDATE, TIMESTAMP_NTZ, zero nulls across 4,314,292 / 1,073,135 / 108,615,680 rows	custom
Operation code	__OPERATION_TYPE present, but only L and A observed and no delete code	none generated; SAP business deletion codes present and populated (section 5)	neither
Load phase	load_type declared	inferable - 4,314,285 rows on the load day, then 6, then 1	standard
Flow identity	run_id, measured not to be a batch discriminator	none	neither

Capability	Standard product	Custom product	Ahead
Predicate pruning	NOT demonstrated - the reference objects are a single micropartition (39,462 rows, 320 KB) and the predicate read essentially everything	0.00 MB against 7,035.80 MB for a full read - 0.0000% - on a 108,615,680-row object	custom
Free no-change poll	0.000 MB in 2 ms, but only if compared as text	0.00 MB in 47-79 ms	tie

The engine is identical on both tracks and proven on both. On the custom track an initial refresh inserted 4,314,292 rows in 18.3 s and the next no-change refresh returned NO_DATA in 984 ms, with 10 of 10 Gold-layer dynamic tables INCREMENTAL once REFRESH_MODE was pinned explicitly. On the standard track a dynamic table over the live share refreshed INCREMENTAL in 5,241 ms and then returned NO_DATA in 610 ms, with a watermark poll returning REFRESHED and then NO CHANGE.

REFRESH_MODE must be stated explicitly on both tracks. Left to AUTO, Snowflake selects FULL on a non-trivial query, and FULL cascades - change tracking is unsupported on a FULL dynamic table, so everything downstream is forced to FULL as well.

5. The finding that reorders the priorities

The custom track has no connector-generated operation column, and the usual conclusion drawn from that is that deletes are simply invisible. That conclusion is too broad. SAP's own business deletion and blocking codes were present on the reference custom share, populated, and material - and the Gold layer dropped them.

Deletion code on purchase order items	Items	Share
(blank) - active	3,494,547	81.0%
L	812,913	18.8%
S	6,831	0.2%
X	1	0.0%
Flagged in total	819,745	19.0%

Restricted to a single currency, so that the figures are additive: of 69.50 bn USD of ordered value on those items, 40.17 bn sits on active items and 29.33 bn - 42.2% - sits on items carrying a deletion or blocking code. The same 69.50 bn is the headline ordered-value figure the Gold layer reports.

> The deletion codes exist in the Silver layer and do not reach Gold. The Gold purchase order table carries all 4,314,292 rows, the same count as its source, with no deletion column and no filter. Deleted and blocked items are therefore included in reported value, unlabelled and unfilterable by the consumer.

The same pattern appears elsewhere on the share: a deletion code on purchase order headers (1,594 flagged), ISMARKEDFORDELETION on products (1,086 flagged), and DELETIONINDICATOR, ACCOUNTISMARKEDFORDELETION and SUPPLIERCENTRALDELETIONISBLOCK across master data.

Two qualifications matter. These are BUSINESS deletion flags - a record marked deleted in the source - not replication operation codes; a row physically removed from the share still leaves no trace on this track. And

whether L and S items should be excluded from spend, shown separately, or retained is a business decision rather than a technical one. What is not defensible is carrying them silently and unlabelled, which is the current behaviour.

6. The recommendation

In priority order, highest value first. Only item 3 requires anything from SAP.

6.1 Surface the change metadata the share already carries

Propagate the deletion and blocking codes from Silver to Gold and into the semantic layer, so a consumer can filter or label them. This costs nothing, requires no SAP change, and on the reference account it governs 19.0% of purchase order items and 42.2% of reported USD ordered value. It also delivers most of what an operation column would deliver, because logical deletion is the dominant deletion mode in SAP.

6.2 Keep the watermark strongly typed - do not copy the standard form

Specify the change timestamp as TIMESTAMP_NTZ and NOT NULL. The reference custom product already achieves zero nulls across 114 million rows; make that a contract rather than a happy accident. Explicitly do NOT adopt __TIMESTAMP's TEXT-with-comma representation - it casts silently to NULL and would be a regression.

6.3 Ask for a replication operation code, or a soft-delete guarantee

This is the one genuine capability gap, and the only item here that is an ask to the pipeline owner. Either emit an operation code that distinguishes insert, update and physical delete, or guarantee that rows are soft-deleted rather than removed. A row that simply stops being carried is unrecoverable by any watermark design. Note that the standard track has not been shown to satisfy this either, so it is an ask for both tracks rather than a reason to prefer one.

6.4 Add a declared load-phase flag

A column distinguishing the initial snapshot from subsequent deltas removes the need to infer the load phase from the timestamp distribution. Inference works today - 4,314,285 rows on the load day, then 6, then 1 - so this is a convenience rather than a requirement.

6.5 Do not replicate run_id

It was measured to hold the same value on the initial snapshot and on every delta, so it identifies the flow rather than the batch and adds no capture capability. Copying it into a custom product would add a column and no information.

7. What this means for choosing between product types

Where a standard data product exists, prefer it - for one structural reason. It carries an operation column at all, which means a delete can in principle be represented, and no source-system configuration is required to obtain it. That is a real advantage and it is the right default.

But a standard product is not a shortcut to a correct design. It ships a worse timestamp representation, an operation enumeration that could not be confirmed and in which no delete code was ever observed, a flow identifier of no use to capture, and metadata that is destroyed by the ordinary act of copying the share. Its predicate pruning was not demonstrated. A well-configured custom product with its deletion codes surfaced is materially closer to a correct and economical CDC design than an unvalidated standard product is.

The practical conclusion is that product type is a weaker determinant of CDC quality than configuration and validation are. Both tracks run the same engine, and on both the deciding questions are whether the watermark is trustworthy, whether the connector merges rather than reloads, and whether deletion is represented at all.

8. What is not established

Four things are open, and each is stated as open rather than smoothed over:

First, the authoritative __OPERATION_TYPE enumeration, and whether a delete code is ever emitted. Only L and A were observed. The SAP documentation page is JavaScript-rendered and could not be read; a web search surfaces the SAP Datasphere BigQuery adapter, which is a different target and is deliberately not cited here as authority. Until this is answered, treat physical deletes as unrepresented on both tracks and reconcile row counts on a schedule.

Second, whether a watermark predicate prunes on the standard track at production scale. It did not prune on the reference objects, but those are a single micropartition, so the test had no headroom.

Third, what a dynamic table refresh does when the source has genuinely changed after the initial load. Refreshes were observed as INCREMENTAL on first build and NO_DATA when nothing had changed; a refresh driven by a new connector write was not observed on either account.

Fourth, whether the SAP deletion codes L, S and X should exclude items from spend reporting. That is a business determination, not a measurement, and this paper deliberately stops at establishing that the codes are populated, material, and currently discarded.

9. Verifying this on your own tenant

None of the above should be taken on trust. The accompanying probe classifies every object by track, validates the standard metadata columns, measures poll and extraction cost separately, and returns a verdict:

```
cd probes
python3 cdc_readiness.py --connection <your_account> --report readiness.md
```

The validation step matters most, because the metadata columns cannot be trusted on sight. On the reference tenant they were clean inside the live catalog-linked database and corrupted on every local copy of it - on one product __OPERATION_TYPE held media titles, and on another __OPERATION_TYPE, __TIMESTAMP and LOAD_TYPE all held the same customer identifier on 735,305 of 735,305 rows with RUN_ID null. The thresholds the probe applies come from that measured contrast: an operation column with cardinality above 12 or values longer than two characters is not a change code, a timestamp column where nothing parses is unusable, and operation equal to timestamp on any row indicates a damaged copy.

The deployable SQL is in sql/, with sql/05_standard_product_track.sql covering validation, the free poll, freshness and a merge template for the standard track, and sql/04 covering the custom-track exception path.

Point capture at the catalog-linked database itself and never at a copy of it.